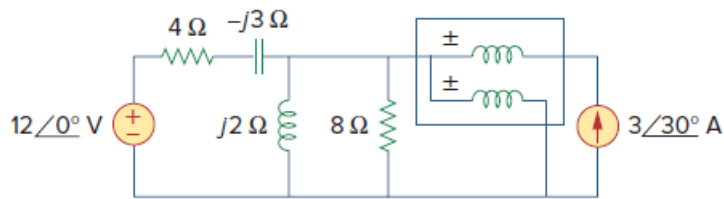


Problem

Obtain the wattmeter reading of the circuit in Fig. 11.91.

Figure 11.91:



Step-by-step solution

Step 1 of 2

The wattmeter reads the real power supplied by the $3\angle 30^\circ$ A current source.

Let V_1 be the voltage across the $3\angle 30^\circ$ A current source.

Redraw the circuit as shown in Figure 1.

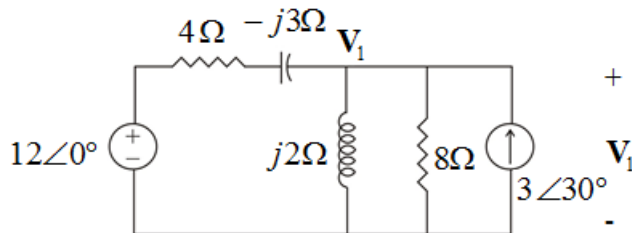


Figure 1

Apply nodal analysis at node V_1 .

$$\frac{V_1 - 12}{4 - j3} + \frac{V_1}{j2} + \frac{V_1}{8} - 3\angle 30^\circ = 0$$

$$V_1 \left[\frac{1}{4 - j3} + \frac{1}{j2} + \frac{1}{8} \right] = 3\angle 30^\circ + \frac{12}{4 - j3}$$

$$V_1 [0.16 + j0.12 - j0.5 + 0.125] = (2.6 + j1.5) + (1.92 + j1.44)$$

$$V_1 [0.285 - j0.38] = (4.52 + j2.94)$$

Simplify further.

$$\begin{aligned} V_1 &= \frac{4.52 + j2.94}{0.285 - j0.38} \\ &= 0.757 + 11.32j \\ &= 11.347\angle 86.19^\circ \text{ V} \end{aligned}$$

Thus, the voltage across the current source is $11.347\angle 86.19^\circ$ V.

Step 2 of 2

Calculate the complex power.

$$S = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^*$$

Substitute $11.347 \angle 86.19^\circ \text{ V}$ for $\mathbf{V}_{\text{rms}}$, $3 \angle 30^\circ \text{ A}$ for $\mathbf{I}_{\text{rms}}$.

$$\begin{aligned} S &= 11.347 \angle 86.19^\circ \times (3 \angle 30^\circ)^* \\ &= 11.347 \angle 86.19^\circ \times (3 \angle -30^\circ) \\ &= 34.041 \angle 56.19^\circ \\ &= 18.94 + j28.28 \text{ VA} \end{aligned}$$

Calculate the wattmeter reading.

$$\begin{aligned} P &= \text{Re}(S) \\ &= \text{Re}(18.94 + j28.28) \\ &= 18.94 \text{ W} \end{aligned}$$

Therefore, the wattmeter reading is 18.94 W.